

Lecture 26: Stochastic Processes (Part 1)

Strict Sense Stationary (SSS), Wide Sense Stationary (WSS), and Ergodic Process

1 Basics of Stochastic Processes (Revision)

1.1 Collection of Waveforms

Definition (Stochastic Process)

A stochastic process is defined as:

$$X(\zeta, t)$$

Logical Interpretation

An entire collection of waveforms or functions over time dependent on the outcome of a random experiment.

The definition emphasizes that:

- The process is not a single function but a collection of possible functions (waveforms).
- Each waveform corresponds to an outcome of a random experiment.
- Therefore, the output is inherently random.

Key Observation

The random output heavily depends on the experiment.

Example (From Slides)

Tossing a coin:

- Outcomes: Heads or Tails
- Each outcome produces a specific deterministic waveform

Hence:

Random experiment $\rightarrow$ deterministic realization

1.2 Realizations and Random Variables

Definition (Realization)

A specific measured waveform from the experiment is called a realization of the process.

Fixed-in-Time Representation

Step 1: Fix Time

Consider a specific time $t = t_1$

Step 2: Evaluate Process

The stochastic process reduces to a single random variable:

$$X(\zeta, t_1) = X(t_1)$$

Step 3: Extend to Multiple Times

At multiple time instants (t_1, t_2) :

$$X(t_1), X(t_2)$$

These are multiple dependent random variables.

Key Conclusion

The dependency between these random variables determines:

2 Characterization of Stochastic Process

2.1 Probability Density Functions (PDF)

1st Order Characterization

Describes the random variable at a single time t_1 :

$$f_X(x, t_1), \quad F_X(x, t_1)$$

Reasoning

Since fixing time gives a random variable:

→ Its behavior is captured using PDF/CDF

2nd Order Characterization

Describes joint probability at two time instants (t_1, t_2) :

$$f_{X_1, X_2}(x_1, x_2, t_1, t_2)$$

Reasoning

Since multiple time evaluations give multiple dependent random variables:

→ Their joint distribution is required

nth Order Characterization

Complete statistical description:

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n, t_1, \dots, t_n)$$

Conclusion

Increasing order → more complete description of the process

3 Statistical Moments

The fundamental nature of the stochastic process

3.1 Mean of Stochastic Process (First Order Moment)

Definition

The mean of a stochastic process is:

$$\mu_X(t) = E[X(t)]$$

Step-by-Step Computation

Using first-order PDF:

$$\mu_X(t) = \int_{-\infty}^{\infty} x \cdot f_X(x, t) dx$$

Observation

The mean is generally a function of time:

$$\mu_X(t) \neq \text{constant}$$

Reasoning

Since the distribution can vary with time:

→ The expected value also varies with time

3.2 Autocorrelation Function

Purpose

Measures dependence between values at different time instants

Definition

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

Interpretation

Captures relationship between:

$$X(t_1) \quad \text{and} \quad X(t_2)$$

Logical Development

1. Multiple time evaluations → multiple random variables
2. These variables are dependent
3. Their dependency is quantified by autocorrelation

4 Conceptual Flow of the Lecture

The lecture builds logically as follows:

1. Start with definition
Stochastic process = collection of waveforms
2. Fix time
Leads to random variables
3. Multiple time instants
Leads to dependent random variables
4. Characterization
Use PDFs (1st, 2nd, nth order)
5. Statistical summaries
Mean $\rightarrow$ first-order behavior
Autocorrelation $\rightarrow$ dependency structure

5 Summary for Exam Revision

- A stochastic process is a collection of time functions indexed by random outcomes
- Fixing time $\rightarrow$ random variable
- Multiple times $\rightarrow$ dependent random variables
- Full description requires nth-order PDFs

Mean:

$$\mu_X(t) = E[X(t)]$$

Autocorrelation:

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$